ABHAY SINGH

Python Developer | BCA Student

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PROFESSIONAL SUMMARY

BCA student currently in the 5th semester with a strong foundation in Python, problem-solving, REST API integration, JSON handling, backend development, and Machine Learning. Built practical projects including AI chatbots, automation tools, a weather application, an expense tracker, and a Height & Weight prediction system using Linear Regression. Currently working as a backend developer in a 5-member team on an education portal using Python, Django, MySQL, and REST APIs. Seeking internship or entry-level opportunities in Python, backend, or AI application development.

TECHNICAL SKILLS

Programming: Python, HTML, CSS, JavaScript (Basics)
Backend & APIs: Django, REST API Integration, JSON Data Handling
Machine Learning: Scikit-learn, Linear Regression, Pandas, NumPy, Model Evaluation (R^2, MAE, RMSE)
AI & LLM: LLM API Integration (Groq / Llama 3.1), Prompt-Based Application Design
Tools & Platforms: Git, GitHub, VS Code, Streamlit, Streamlit Cloud, GitHub Pages
Other: File I/O, Environment Variables, Secrets Management, OpenCV, MediaPipe | IoT (Basics): Sensors, Actuators, Arduino/ESP32, GPIO, MQTT

EDUCATION

Bachelor of Computer Applications (BCA) — Infopark – The College of IT & MGT
Currently in 5th Semester | 2024 – 2027

EXPERIENCE

Backend Developer (Team Project) – Education Portal — June 2026 – Present | In Progress

- Working as a Backend Developer in a 5-member team building a real-time education portal.
- Designing and developing REST APIs, database operations, authentication, and server-side logic.
- Working with Python, Django, MySQL, REST API, HTML, CSS, and JavaScript.
- Collaborating with the frontend team and supporting reliable data flow and application performance.

PROJECTS

Height & Weight Predictor | Python, Scikit-learn, Streamlit, Pandas, NumPy

- Built a Linear Regression model to predict height and weight based on age using real children's dataset (Kids198).
- Performed data cleaning, feature engineering (unit conversion), train-test split, and model evaluation (R^2, MAE, RMSE).
- Developed and deployed an interactive Streamlit web app for real-time predictions with visualization.

Education Portal (Real-Time Project) | Python, Django, MySQL, REST API, HTML, CSS, JavaScript

- Currently working in a 5-member team to build a full-featured education portal.
- Responsible for backend development, REST APIs, database handling, authentication, and server-side logic.

Smart Hand Gesture Controller | Python, OpenCV, MediaPipe

- Developed a real-time hand gesture system using OpenCV and MediaPipe.
- Controls mouse functions such as move, click, drag, scroll, and desktop switching with smart gestures.
- Added features including volume control and window management.

AI Chatbot | Streamlit, Groq API, Llama 3.1

- Built and deployed a live conversational chatbot powered by Groq API (Llama 3.1).
- Implemented a persistent chat interface in Streamlit with secure API key handling.

Neural Habit Grid | Streamlit

- Built a sci-fi themed habit tracker with self-rated daily check-ins.
- Implemented automatic streak calculation and a dynamic UI with CSS styling and animations.

Weather App | Python, OpenWeatherMap API

- Fetches and displays live weather data for any city using a REST API.
- Practiced secure API-key handling using Python dotenv.

Expense Tracker | Python CLI

- Built a menu-driven command-line application to add, view, and categorize expenses.
- Uses local storage for persistent expense data.

CERTIFICATIONS

Certificate of Participation — Master Class on AI Tools for Students
BrainByte Solutions | 19 September 2026 | Certificate ID: BBS-AI-2026-120

ADDITIONAL SKILLS

Video Editing (Basic): CapCut, VN, DaVinci Resolve | Frontend UI Design: HTML, CSS, Basic JavaScript | Figma (Beginner)